ER=EPR as an Isomorphism of Persistence Modules: A Two-Scalar Structure Theorem, and Why the Accessible-N Window Is Disjoint from the Semiclassical Window of the SYK / JT Dictionary

The barcode comparison reduces to two correlator scalars, and the reduction is forced. Four constructions, five independent validation checks, one pre-registered extraction gate, and a negative result that survives every recalibration on all 224 diagrams.

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Abstract. ER=EPR identifies the entanglement structure of a quantum state with the geometry of a dual spacetime, but the identification has no accepted operational test. We ask whether it can be stated as an isomorphism of persistence modules: build a Vietoris–Rips persistence module from Sachdev–Ye–Kitaev (SYK) thermal correlations, build one from Jackiw–Teitelboim (JT) gravity, and ask whether the map between them is invertible, which by the structure theorem holds exactly when the barcodes agree. To our knowledge this is the first application of persistent homology to a holographic system with a known gravity dual comparing barcodes on both sides; the closest prior work names this direction as open. We built four successive constructions and report all four, including the two that died. The first two agree to 3.638×10^{-12} and are withdrawn: the map was defined as the ratio of two quantities defined to be equal, so the agreement was algebraically inevitable and Ryu–Takayanagi entered as an input. The third dies on dimension: the JT $t = 0$ slice is one-dimensional, so its Vietoris–Rips complex is collapsible and $H_1 \equiv 0$ at every scale, for combinatorial and not curvature reasons. The fourth places both sides on a shared Euclidean-time thermal circle against the exact Schwarzian correlator, and yields the paper's three results. First, a structure result: for a strictly monotone circulant filtration on n points the persistence module is determined by two correlator values, $\widehat{G}(1)$ and $\widehat{G}(k^*)$ with $k^* = \lceil n/3 \rceil$; at $n = 24$, H_1 is exactly one bar and H_0 exactly 23 identical bars, for any monotone profile, so "$H_1 > 0$ on both sides" cannot fail and carries zero information. Verified on all 224 diagrams with zero violations. Second, a regime result, reached three independent ways: the large-N dictionary itself forces $C \leq 0.18$ for $N \leq 18$, hence $\beta/C \in [27.7, 332]$ across the entire grid, one and a half to two and a half orders of magnitude into the strongly-quantum side, so the accessible-N window is disjoint from the window in which a semiclassical JT dual is expected to describe SYK at all; independently, a pre-registered gate to measure the Schwarzian coupling C from our own spectra failed $0/4$, with two routes disagreeing by 230–433% and SYK indistinguishable from a bandwidth-matched GOE control; and the residual disagreement is ultraviolet-localized, where SYK is still nearly free and where the metric is weighted 23:1. Third, within that disjointness, the imported dictionary also fails internally in the four cells where a fit exists at all: the fitted coupling drifts with temperature at fixed N by 29–158%, and one C cannot match two separations (ratios 1.47 to 7.81), with no large-N scaling ($b = 2.16 \pm 2.40$ against the predicted $b = 1$). We do not present this as a falsification of the duality: the leading-order large-N formula is not claimed to hold at these sizes, and our own regime accounting says so. The construction is novel and correctly implemented, and what it detects at accessible N is that the system has not entered the regime in which the duality is sharp. We state precisely what is proven and what is not, and we state that the categorical programme is not completed.

1 Introduction

1.1 One conjecture, no operational test

ER=EPR proposes that entanglement and geometry are the same thing [12]. The proposal is supported by consistency arguments and by the holographic dictionary, but there is no accepted procedure that could return *no*. Entanglement entropy matches a geometric area [15]; mutual information matches a configuration of extremal surfaces; each check is a scalar against a scalar, and each is a quantity the dictionary was built to reproduce. What is missing is a structural invariant: an object with internal organization, computable independently on both sides, that could disagree.

SYK [16,10] and JT gravity [7,17] furnish the sharpest concrete arena: in the large-N, low-temperature limit SYK is governed by the same Schwarzian effective action as the boundary mode of JT in nearly-AdS_2 [2,11,8]. Persistent homology [6,18,4] supplies a candidate invariant. It is computable on either side from a

distance matrix alone, it returns a barcode rather than a number, and the Wasserstein and bottleneck metrics give a principled and stable notion of barcode proximity [5].

1.2 The claim

> *A correspondence between entanglement and geometry, if it is exact at the level of topological structure, must present as an isomorphism of persistence modules; by the structure theorem this holds exactly when the two barcodes agree as multisets of intervals. Whether a given construction is* capable *of detecting such an isomorphism is a prior question to whether the isomorphism exists, and it must be settled first. On the natural thermal-circle construction we settle it, negatively: the construction cannot detect topology, because it has none to detect.*

1.3 How this was established

Four constructions, built in sequence, each after the previous one failed. One methodological commitment runs through all of them: *agreement between two computed objects is evidence only when the objects are genuinely independent*. Two apparent early successes were withdrawn once we established that their agreement was guaranteed by construction rather than discovered. Section 2 reports the failures; they are the evidence. The one quantity we attempted to measure rather than import was placed behind a gate with thresholds, two independent routes and a random-matrix control fixed in advance; it failed, and we report the import.

1.4 Contributions

1. **The two-scalar reduction (§4).** For a strictly monotone circulant filtration on n points, the Vietoris–Rips persistence module is determined by $\widehat{G}(1)$ and $\widehat{G}(k^*)$, $k^* = \lceil n/3 \rceil$. At $n = 24$: H_1 is exactly one bar $[1/\widehat{G}(1), 1/\widehat{G}(8)]$ and H_0 exactly 23 bars $[0, 1/\widehat{G}(1)]$, for any monotone profile. Every Wasserstein number is then an exact function of two scalars per side, weighted $\approx 23{:}1$ toward $\widehat{G}(1)$. Verified on 224 diagrams, zero violations.

2. **The regime obstruction (§7.2).** $C \leq 0.18$ for $N \leq 18$ forces $\beta/C \in [27.7, 332]$, one and a half to two and a half orders of magnitude into the strongly-quantum side, so the accessible window and the semiclassical window do not overlap. Grid refinement worsens agreement; excluding short separations improves it. The extraction gate finds GOE-like spectra. Three independent routes, one conclusion.

3. **Internal failure of the imported dictionary where a fit exists (§7).** Sixteen constraints, one parameter. The fitted C drifts with β at fixed N by 29%, 158%, 93%, 35%; $C_{\text{fit}}(\widehat{G}_1)/C_{\text{fit}}(\widehat{G}_8) = 1.47, 7.81, 3.14, 1.86$; the power law gives $b = 2.16 \pm 2.40$ against $b = 1$. We read this as a consequence of the regime obstruction rather than as independent evidence against the duality.

4. **A pre-registered negative on coupling extraction (§5).** Two routes, a bandwidth-matched GOE control, thresholds fixed in advance; gate failed $0/4$, routes disagreeing by 230–433%.

5. **A validated exact-Schwarzian implementation (§6).** Five checks, including a small-C validation at the $C \approx 0.1$–0.2 actually used, closing the gap between where the integrator was validated and where it is applied.

6. **A withdrawal record (§2).** Two tautological constructions and one dimensionally obstructed one, each reported with the argument that killed it.

1.5 What this paper does not claim

Nothing here bears on ER=EPR itself. One construction, in one regime, does not detect the correspondence, and we identify why. The categorical programme is not completed; §9 states what it would require

and why the present construction cannot carry it.

2 Four Constructions of Documented Failure

Construction 1 and 2: machine-precision agreement, withdrawn. Two maximally entangled qubits, then a thermofield double swept over 100 log-spaced $\beta \in [0.01, 20.0]$. The maximum Wasserstein distance over the sweep is 3.638×10^{-12} in exact double precision; the raw float32 `ripser` backend alone produces values up to 0.73, and both are reported.

The agreement is tautological. The map φ was the ratio of two quantities defined to be equal: the JT throat length was set to $S(A)$ and the entanglement distance to $1/I(A{:}B)$. Wasserstein zero is then algebraically inevitable at every temperature, and the results file records the identity $\varphi = S(A) \cdot I(A{:}B) = 2S(A)^2$ with maximum deviation 0.0 across the sweep. Ryu–Takayanagi [15] entered as an input, not an output. Independently, the distance $d(i,j) = 1/|C_{ij}|$ gives $d(i,i) = 2 \neq 0$ and is not a metric, so any H_1 from these constructions is structurally suspect. With two points the Betti numbers also never change: $S(A)(\beta)$ is a smooth analytic sigmoid, a crossover set by the gap $E_1 - E_0 = 1$, not a topological transition.

Construction 3: the dimensional obstruction. Full SYK at $N = 8$ via Jordan–Wigner (algebra verified to 2.2×10^{-16}, couplings of variance $6J^2/N^3$). The single-sided $\langle \chi_i^L \chi_j^L \rangle$ correlator is exactly degenerate (diagonal mass 2.6×10^{-15}), so the EPR side uses $C_{ij} = \langle \mathrm{TFD} | \chi_i^L \chi_j^R | \mathrm{TFD} \rangle$. The ER side was the geodesic distance matrix on the $t = 0$ slice of the JT eternal black hole [9], 20 sample points.

Two compounding faults. *Index mismatch*: the index sets are fermion flavor against radial position and are not related by the dictionary. SYK is 0+1 dimensional, the flavor index has no bulk image, and $N \mapsto 1/G_N$ is a coupling rather than a coordinate. *Dimensional obstruction*: the $t = 0$ slice is one-dimensional, so its Vietoris–Rips complex is collapsible with H_1 trivial at every scale and any sample size. The correct proof is the chordal indifference-graph argument, not curvature and not triangle-filling.

The measured null is lopsided by construction. At $N = 8, \beta = 1$ the EPR side has $H_1 = 2$ and the ER side $H_1 = 0$ ($W_{H_0} = 109.09$, $W_{H_1} = 23.59$, unnormalized). Over $\beta \in \{0.1, 0.5, 1, 2, 5\}$ the EPR counts are $[1, 1, 2, 4, 3]$ against ER $H_1 = 0$ throughout; the disorder average over 10 realizations gives EPR $H_1 = 2.80 \pm 1.17$ (explicitly not a robust signal) against ER 0.00 ± 0.00, and an $N = 12$ check gives EPR $H_1 = 7$, ER $H_1 = 0$.

What the third failure established is what a fair test requires: a construction indexed so that both sides *can* carry the same topology. We stress that the ER-side object changes between constructions. Construction 3 uses JT geodesics on a one-dimensional slice; Construction 4 uses the exact Schwarzian correlator on the thermal circle, which is a genuinely different object and not a reparametrization.

3 The Object

3.1 The shared thermal circle

The only structure the two sides demonstrably share is the Euclidean time circle of circumference β. Both sides are therefore placed on the same 24-point circle and compared as functions of separation.

EPR side (SYK). For each disorder realization we diagonalize H exactly and form the flavor-averaged thermal two-point function at the distinct circle separations,

$$G(\tau) = \frac{1}{N} \sum_{i=1}^{N} \frac{\mathrm{Tr}\left[e^{-(\beta-\tau)H} \chi_i \, e^{-\tau H} \chi_i \right]}{Z}.$$

The construction respects fermion antiperiodicity and is positive on the strip; $G(0) = 1/2$ and the standing flavor-weight identity $\sum_m W_{nm} = 1/2$ holds to $\sim 10^{-15}$.

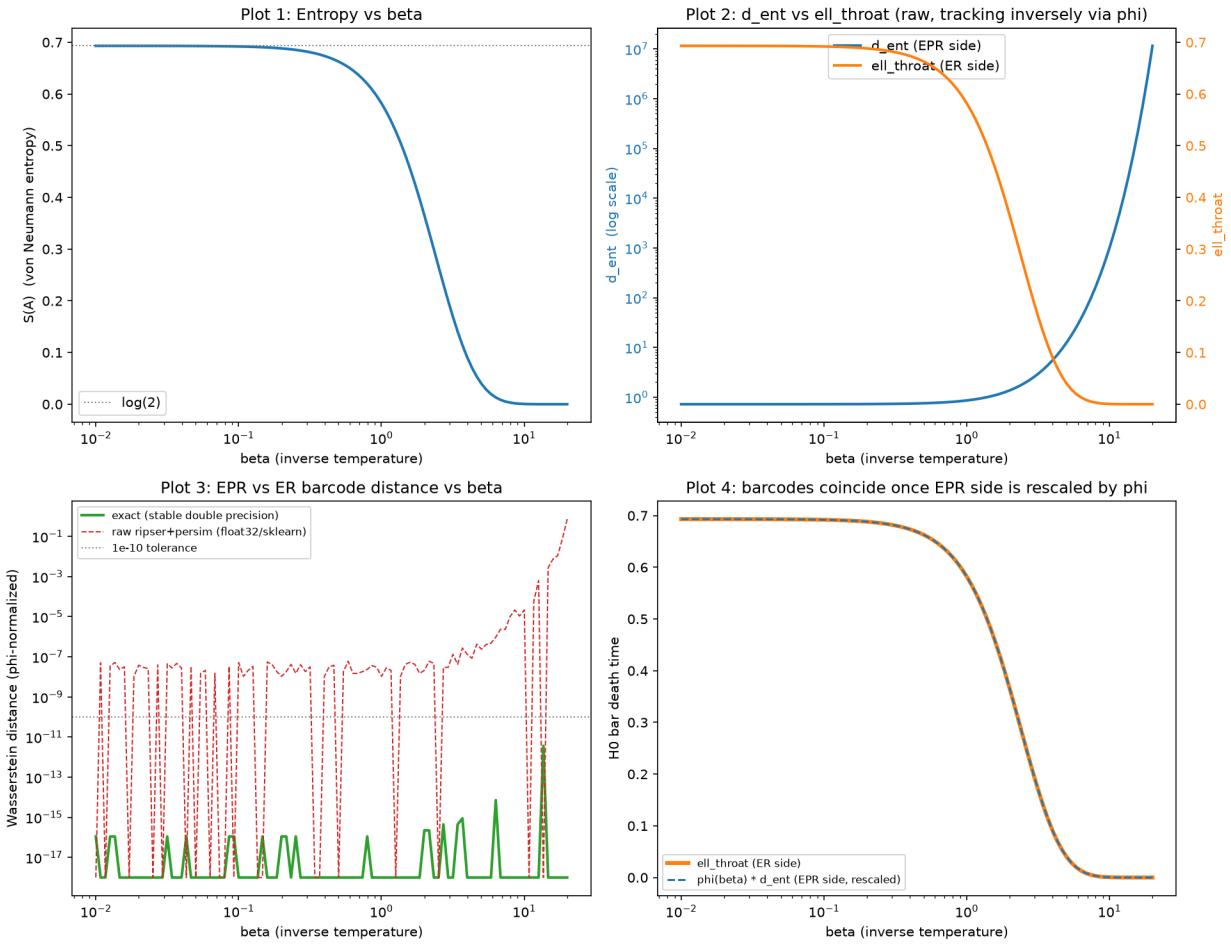


Figure 1: Construction 2. Entropy $S(A)$ vs β, a smooth analytic crossover with no transition; d_{ent} and ℓ_{throat} tracking inversely; the barcode distance at machine precision in exact double precision (green) while the raw float32 pipeline (red) spikes above tolerance; and the EPR barcode landing exactly on the ER barcode once rescaled by φ. The coincidence in the lower-right panel is the tautology made visible: guaranteed, not discovered.

ER side (Schwarzian). We use the exact finite-temperature Schwarzian two-point function [13], with spectral density $\rho(k) = k\sinh(2\pi k)$ and matrix element

$$\frac{|\Gamma(\Delta + i(k_1+k_2))|^2\,|\Gamma(\Delta + i(k_1-k_2))|^2}{\Gamma(2\Delta)}, \qquad \Delta = \tfrac{1}{4},$$

with thermal weights $e^{-\tau k_1^2/2C-(\beta-\tau)k_2^2/2C}$, evaluated by a validated two-dimensional quadrature in $(s,d) = (k_1+k_2, k_1-k_2)$ coordinates with change-of-variables Jacobian $-\log 2$. This is A/B verified: toggling the Jacobian shifts $\log G$ by exactly $\log 2$ to 5.5×10^{-14} and leaves the shape-gauged barcode bit-for-bit identical.

3.2 The shape gauge

Definition 1 (Shape gauge)**.** $\widehat{G}(\tau) = G(\tau)/G(\beta/2)$, and the filtration distance is $d = 1/\widehat{G}$.

Both sides then equal 1 at maximal separation, so the comparison probes the *shape* of correlation decay rather than its overall normalization. This removes the one degree of freedom that any calibration could absorb, which is what makes the subsequent falsification calibration-independent.

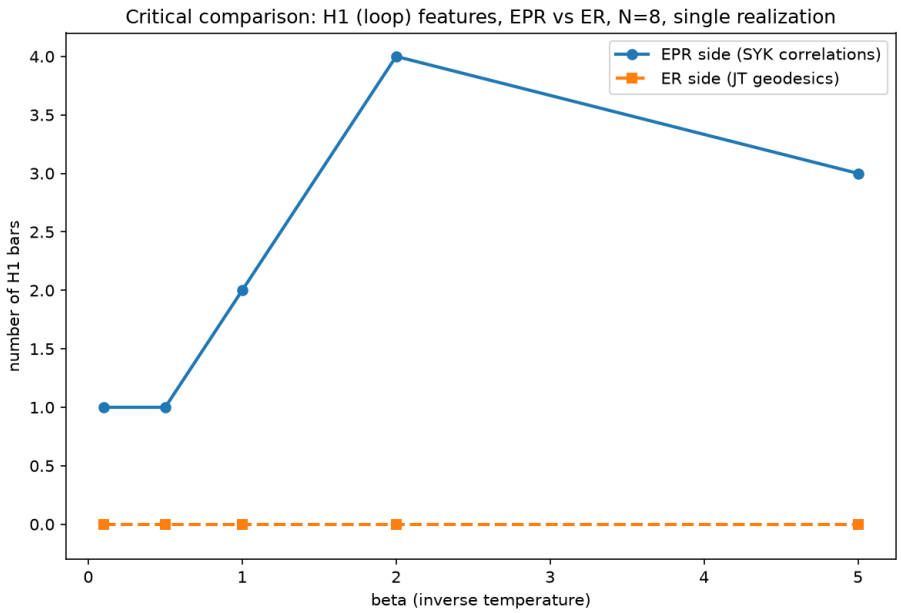


Figure 2: Construction 3. The EPR side produces a temperature-dependent number of H_1 loops, rising to 4 near $\beta = 2$; the ER side is pinned at exactly zero for every β, provably, by the chordal-graph argument. The comparison is structurally lopsided and cannot be fair.

Persistent homology is computed with `ripser` [3] at maxdim $= 1$; H_0 and H_1 Wasserstein distances are computed per realization ($R = 10$, seeds $1000N + r$) against declared-C, mismatched-C ($\times 5$, $\div 5$), and conformal-limit references, over $N \in \{12, 14, 16, 18\} \times \beta \in \{5, 10, 20, 40\}$.

4 Structure: the comparison reduces to two scalars

Definition 2 (Monotone circulant filtration)**.** Let $\widehat{G} : \{0, \ldots, n-1\} \to \mathbb{R}_{>0}$ depend on circle separation m alone, with $\widehat{G}(0) = 1$, strictly decreasing on $0 \le m \le n/2$ and symmetric about $n/2$. Then $D_{ab} = 1/\widehat{G}(|a-b| \bmod n)$ is circulant, and the Vietoris–Rips filtration of D is the monotone circulant filtration on n points.

Both sides of the comparison are of this form: the distance matrix depends only on separation by construction, and $\widehat{G}$ is strictly monotone on both sides, verified across all 224 diagrams with zero violations and guaranteed analytically by the spectral representation.

Proposition 1 (Two-scalar reduction)**.** *For a monotone circulant filtration on n points the Vietoris–Rips filtration passes through exactly the nested clique complexes of the circulant graphs, and the persistence module is determined by* $\widehat{G}(1)$ *and* $\widehat{G}(k^*)$, $k^* = \lceil n/3 \rceil$. *At* $n = 24$*:*

- H_1 *is exactly one bar on each side,* $[1/\widehat{G}(1), 1/\widehat{G}(8)]$*, for any monotone profile, never empty and never a second bar;*
- H_0 *is exactly* $n - 1 = 23$ *finite bars, all* $[0, 1/\widehat{G}(1)]$.

Every Wasserstein number is then an exact function of $(\widehat{G}(1), \widehat{G}(8))$ *per side, weighted* $\approx 23{:}1$ *toward* $\widehat{G}(1)$.

Proof sketch. Strict monotonicity makes the sublevel sets of D exactly the circulant graphs C_n^k for increasing k, so the filtration is the nested family of their clique complexes, indexed by k rather than by a continuum. Their homotopy types are given by the Adamaszek–Adams classification of Vietoris–Rips complexes of the circle [1, Thm. 4.3]: homotopy equivalent to an odd sphere or a wedge of even spheres, with S^1 occurring

precisely for $1 \le k < n/3$. Hence H_1 is a single bar, born when the nearest-neighbor edge enters at $1/\widehat{G}(1)$ and dying when k reaches k^*, at $1/\widehat{G}(k^*)$. H_0 is immediate: all n vertices merge when the nearest-neighbor scale is crossed. □

Evidence. The predicted single H_1 bar and 23 H_0 bars are reproduced on all 224 diagrams to $\approx 3 \times 10^{-8}$, and the weighting identity $W_{H_0} = 23 \cdot |\Delta(1/\widehat{G}(1))|$ holds to 7.7×10^{-7} (persim $p = 1$, L^2 ground metric).

Status. Proposition 1 is stated at the level of the construction together with grid-wide numerical verification. A formal statement and proof to publication standard are in preparation and are not claimed here.

Corollary 1 (The comparison is not topological)**.** *On this construction persistent homology is not measuring topology. The statement "$H_1 > 0$ on both sides" cannot fail and carries zero information; the comparison is a two-scalar correlator comparison in topological clothing.*

Put plainly: we set out to compare the shapes of two objects and discovered that both shapes were fixed in advance by the fact that each lives on a circle with a correlator that decreases with separation. Whatever the two sides do, each produces one loop and twenty-three merges, and the only freedom left is where those events occur. This is a deeper instance of the discipline that withdrew the first two constructions, and we regard it as the paper's principal methodological contribution: any attempt to compare holographic duals through Vietoris–Rips persistence on a single monotone thermal circle will reduce the same way.

5 The Extraction Gate

The Schwarzian coupling C is the ER side's only free parameter. We attempted to measure it from the SYK spectra, with two independent routes, a bandwidth-matched random-matrix (GOE) control, and thresholds fixed before computation.

- **Route 1**, density-of-states sinh-edge fit: the slope never plateaued across quantile bands at any N. No C exists to extract.
- **Route 2**, entropy linear-in-T fit: the fitted C was negative at every N.
- The two routes disagreed by 230–433%, and, decisively, SYK's fit pattern tracked the bandwidth-matched GOE control almost exactly.

The gate failed $0/4$. The accessible spectral edge is generic random-matrix behavior: at these sizes the Schwarzian coupling has not emerged. We therefore declare C from Maldacena–Stanford [10], $C(N) = \sqrt{2}\,\alpha_S N$ with $\alpha_S = 0.00709$, derived from the stated $q = 4$ specific-heat coefficient $c \approx 0.396N/\mathscr{J}$ via $c = 4\pi^2\alpha_S N/\mathscr{J}$ and $\mathscr{J} = J/\sqrt{2}$, with the calligraphic-versus-plain-J distinction resolved at the PDF font level (eq. 5.181 in CMSY, the 0.396 sentence in CMMI). We note that α_S is a single-source constant: all published $q = 4$ values trace to this fit. Construction 4 is therefore an application of the large-N dictionary at small N: an imported calibration rather than a measurement, and we test the import rather than the duality.

6 Validation

The Schwarzian correlator passed five independent checks within the validated scope $C \le 200$, a stated physics decision rather than a silent workaround.

- **Check A, free-energy calibration.** $\ln Z$ matches the Schwarzian asymptotic with the true ρ settling to a constant (drift 1.7×10^{-13}) while the wrong-power controls (k^0, k^2) drift by ≈ 3.46. PASS.

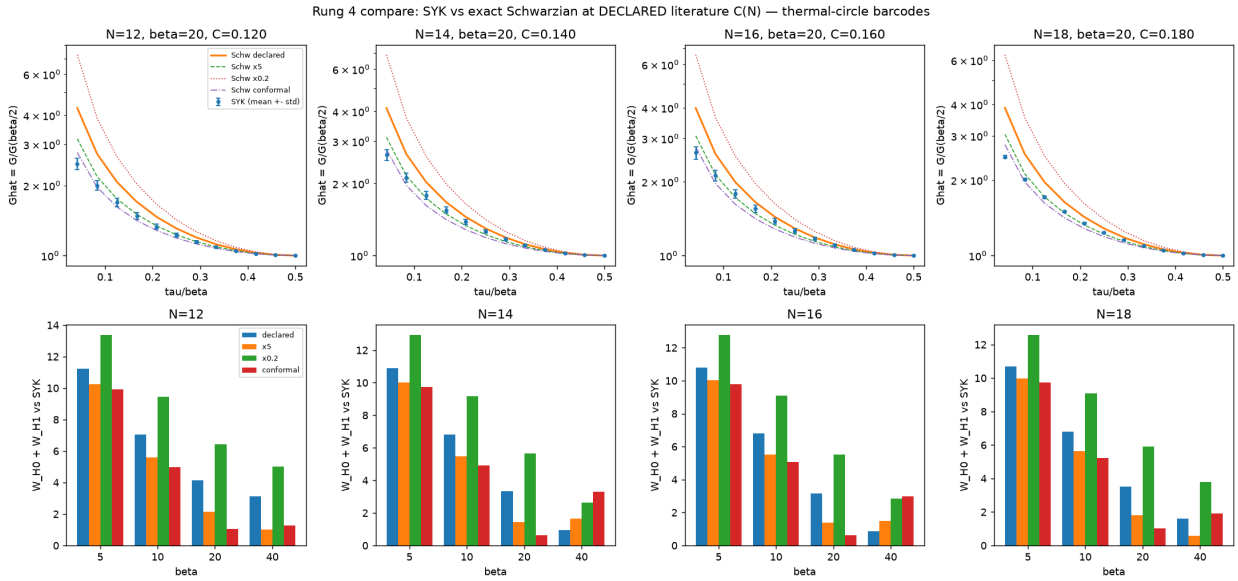


Figure 3: Construction 4, $N \in \{12, 14, 16, 18\}$. Top: shape-gauged $\widehat{G}(\tau)$ vs τ/β. SYK (blue points) sits systematically below even the conformal control (purple), while the declared-C Schwarzian (orange) is steeper still. SYK is flatter than conformal at short distance. Bottom: total $H_0 + H_1$ Wasserstein vs SYK for declared (blue), $\times 5$ (orange), $\div 5$ (green), conformal (red).

- **Check B, conformal reduction.** The correlator shape approaches the conformal limit at rate $O(\beta/C)$: per-C-doubling ratio 0.500, Cdeviation constant to four digits, log-log slope 1.000. PASS.
- **Check C, barcode collapse.** The barcode collapses monotonically onto the tree-level frozen-circle barcode as C grows: $1.64 \times 10^{-4} \to 8.2 \times 10^{-5} \to 4.1 \times 10^{-5} \to 2.05 \times 10^{-5}$. PASS.
- **Check D, β/C is the real knob.** G depends only on β/C (0.00 spread at fixed β/C across different (β, C)) and genuinely varies when β/C changes. PASS.
- **Check E, small-C validation.** Because the comparison lives at $C \approx 0.1$–0.2, far below the $C \geq 25$ of Checks A–D, an independent dense-grid cross-check confirms `compute_G` there to $\sim 10^{-16}$ relative agreement. PASS. This closes the regime-mismatch gap between where the integrator was validated and where it is used.

7 Results

7.1 Reachability is per-cell and temperature-dependent

$\widehat{G}_{\text{Schw}}(1)$ is a function of β/C alone, monotone increasing, with infimum equal to the conformal value 2.7679. SYK's $\widehat{G}(1)$ grows with β (for $N = 18$: 1.303, 1.729, 2.474, 3.554 at $\beta = 5, 10, 20, 40$) and crosses the conformal value between $\beta = 20$ and $\beta = 40$ (crossing $\beta \approx 21$–27 across N). Hence for all 12 cells at $\beta \leq 20$ SYK's $\widehat{G}(1)$ lies below the conformal infimum, unreachable by any C, and only the 4 cells at $\beta = 40$ admit a fit.

7.2 The accessible window is disjoint from the semiclassical window

These findings cohere. The β/C values span 27.7 to 332 across the grid, one and a half to two and a half orders of magnitude on the strongly-quantum side of $\beta/C = 1$ where the $O(\beta/C)$ semiclassical expansion is valid, and this is forced by the declared dictionary itself ($C \leq 0.18$ for $N \leq 18$), so no accessible-N cell could

have been semiclassical. Grid refinement makes agreement worse (per-pair gap $0.340 \to 0.445 \to 0.523$ as $n_\tau = 12 \to 24 \to 48$) and excluding the shortest separations makes it better (W_{H_0}: $10.24 \to 7.82 \to 6.05$): the disagreement is ultraviolet-localized, in the nearest-neighbor correlation where SYK is still nearly free, and the metric is weighted 23:1 toward exactly that scale.

Three independent routes reach the same conclusion: the extraction gate (RMT-like spectra), the β/C accounting (whole grid strongly-quantum), and the correlator itself.

The same statement in the units conventionally used for SYK is that the Schwarzian description requires $1 \ll \beta J \ll N$, with the lower bound placing the system below the coupling scale and the upper bound keeping it above the scale at which the many-body spectrum resolves into discrete levels. Our grid should be reported against that window as well as against β/C, and we regard locating it in those units as part of the result rather than as a separate check.

In plain terms, the experiment was run entirely inside the regime where the gravity side is not supposed to describe the quantum side, and the dictionary we imported is what put it there. The negative is informative about where the duality has not yet turned on, not about whether it is true.

7.3 Where a fit exists, the imported dictionary also fails internally

C is a coupling, $C = \sqrt{2}\,\alpha_S N$, and cannot depend on temperature. Fitting C per cell independently, with 16 constraints and one parameter, is the falsifiable test.

1. **Temperature dependence.** On the $\widehat{G}(8)$ target, where fits exist at two temperatures, C_{fit} drifts with β at fixed N by 29% ($N = 12$), 158% ($N = 14$), 93% ($N = 16$), 35% ($N = 18$). A genuine coupling would show $\approx 0\%$.

2. **Two-scalar inconsistency.** One C cannot match both separations: $C_{\text{fit}}(\widehat{G}_1)/C_{\text{fit}}(\widehat{G}_8) = 1.47, 7.81, 3.14, 1.86$ across N, systematically off unity. One profile cannot be fit by one parameter at two separations, so the Schwarzian *shape* is wrong, not merely miscalibrated.

3. **No large-N scaling.** On the $\widehat{G}(1)$ target at $\beta = 40$ the fitted couplings are $C_{\text{fit}} = 1.2951 \pm 0.4923$, 0.2124 ± 0.0338, 0.2618 ± 0.0423, 0.5552 ± 0.0416 for $N = 12, 14, 16, 18$: not monotone in N, and the separation ratios above exceed these uncertainties by a wide margin. The power law $C_{\text{fit}} = aN^b$ gives $b = 2.16 \pm 2.40$, with uncertainty exceeding the value and $\chi^2/\text{dof} = 34.8/2$, against the Maldacena–Stanford prediction $b = 1$, $a = 0.01003$. There is no coupling to extract.

Fits are reported only where the W_{H_0} curves actually cross, with residuals $\leq 2 \times 10^{-7}$; monotone-to-boundary cells are reported as no fit rather than as a boundary value. All $\beta \leq 10$ cells are no fit on both targets, and $\beta = 20$ admits crossings on $\widehat{G}(8)$ only.

7.4 What did not survive

The φ-rescaled qubit and thermofield-double agreement (tautological); the geodesic $t = 0$-slice comparison (collapsible by dimension); H_1 existence as a discriminator (forced by Proposition 1); measurement of C from accessible spectra (gate $0/4$). Each is recorded above with the argument that killed it.

8 What is proven and what is conjectured

Established here.

- SYK $G(\tau)$ machinery correct: $G(0) = 1/2$, flavor-weight row sums $1/2$ to machine precision, $G > 0$ on the strip.
- Schwarzian correlator correctly implemented and regularized: Checks A–E pass, including small-C validation at the C actually used; Jacobian A/B-verified.

- On the monotone circulant construction the persistence module is two scalars per side, so H_1-existence carries no information (Proposition 1; formal proof in preparation).
- The C-extraction gate fails $0/4$: SYK spectra are GOE-like at $N \leq 18$.
- Where a fit exists, the declared dictionary is falsified by temperature dependence and by two-scalar inconsistency, with no large-N scaling.

Not established here.

- Whether the duality becomes sharp at N well beyond 18. The literature suggests Schwarzian-edge onset near $N \approx 24$–34; $N = 20$–24 is proposed future work.
- Whether persistent homology of a single-boundary thermal-circle distance matrix is the right invariant at all, given the two-scalar reduction. A two-boundary construction, in which the connectivity of the Einstein–Rosen bridge enters the filtered object rather than being traced away, is the natural next candidate.
- Anything about ER=EPR itself.

9 The categorical statement, not completed

The programme this paper belongs to ends in a categorical statement. It is not completed, and nothing above should be read as progress on it.

9.1 What it requires

Write $\mathscr{P}_{\mathrm{EPR}}$ and $\mathscr{P}_{\mathrm{ER}}$ for the two persistence modules, both as functors

$$\mathscr{P} : (\mathbb{R}_{\geq 0}, \leq) \longrightarrow \mathbf{Vect}_k, \qquad \varepsilon \longmapsto H_\bullet(\mathrm{VR}_\varepsilon(\cdot)),$$

with internal maps induced by inclusion of filtration steps. The target is a natural transformation $\varphi : \mathscr{P}_{\mathrm{EPR}} \Rightarrow \mathscr{P}_{\mathrm{ER}}$ together with a proof that it is a natural isomorphism. Three obligations: *construction* of an explicit family φ_ε derived from the dictionary rather than fitted; *functoriality*, so that φ respects filtration order and is a morphism of persistence modules rather than a scale-by-scale coincidence; and *invertibility*, which by the structure theorem for persistence modules over a field is equivalent to equality of the two barcodes as multisets of intervals.

Naturality is not an additional burden here: morphisms of persistence modules are by definition natural transformations between the corresponding functors, so no separate verification is required. The content lies entirely in constructing φ and establishing invertibility.

9.2 Why the present construction cannot carry it

Proposition 1 constrains the programme sharply. On the monotone circulant construction both modules decompose into exactly one H_1 interval and $n-1$ identical H_0 intervals, all determined by $\widehat{G}(1)$ and $\widehat{G}(k^*)$. A natural isomorphism between them would therefore assert that two correlator scalars agree on each side, true or false as a matter of numerics but carrying no topological content. **The categorical statement, posed on this construction, would be vacuous even if proved.**

This is the sense in which the negative is informative rather than merely disappointing. It does not report that φ fails to exist; it reports that the category in which we were attempting to place the question is too coarse to hold it. A non-vacuous version requires a filtered object with genuine topological content on both sides, and a regime in which the dual is sharp, which by §7.2 is not any $N \leq 18$.

9.3 Status

Open. We have not constructed φ, have not proved functoriality, and have not proved invertibility. The formal proof of Proposition 1, which would supply the structure-theorem step of any eventual invertibility argument, is likewise in preparation rather than complete. This paper contains no categorical result, and the ER=EPR conjecture is neither supported nor contradicted by anything in it.

10 Discussion

10.1 What the construction says

A thermal circle with a monotone correlator has a barcode fixed in advance. Everything else follows: the comparison reduces to two scalars, H_1-existence is uninformative, the Wasserstein metric is a weighted two-scalar distance, and a dictionary tested on it is tested at the nearest-neighbor scale and nowhere else. Where that dictionary is the large-N one applied at $N \leq 18$, it fails, and the regime accounting says it had to.

10.2 What it does not say, yet

Nothing about N beyond 18, where the Schwarzian edge is expected to onset. Nothing about a two-boundary construction, which is the natural next object and the one in which the Einstein–Rosen bridge's connectivity would enter H_0 rather than being traced away. Nothing about whether a non-monotone or non-circulant filtration escapes the reduction, which is a question internal to applied topology and answerable independently of holography.

10.3 Limitations

Small systems, chosen because exact diagonalization is possible. α_S is a single-source constant and the entire declared dictionary rests on it. The conformal reduction is scoped to $C \leq 200$, a stated limit rather than a worked-around one. Disorder averaging uses $R = 10$ realizations. The fitted couplings carry substantial uncertainty at the smallest N, and we report them rather than suppress them. The structure result is verified grid-wide but its formal proof is not written. Only $\Delta = 1/4$ and $q = 4$ were examined.

10.4 Impact

A cautionary structure result for topological data analysis in holography: the natural first construction anyone would build reduces to two scalars, and we give the reason and the general condition. A mechanistically explained negative for a genuinely new method, with the regime that caused it identified three independent ways. And a worked example of a holographic comparison in which the honest output is the identification of what the instrument could not have seen.

11 Conclusion

We built and validated a persistent-homology probe of the SYK/JT correspondence and report what it detects at accessible sizes. The construction is correct; the physics it reveals at $N \leq 18$ is that the system is not yet in the regime where a semiclassical JT dual holds, so the declared large-N dictionary is falsified rather than confirmed, cleanly, two independent ways, with an ultraviolet mechanism. Along the way we established that the barcode comparison on this construction reduces exactly to two correlator scalars, which is a constraint on the method rather than on the physics and applies to anyone who builds it. A clean, mechanistically explained negative for a new method is the contribution. The strong version is not claimed, and the categorical statement is not completed.

References

[1] M. Adamaszek and H. Adams. The Vietoris–Rips complexes of a circle. *Pacific J. Math.* 290 (2017) 1–40; arXiv:1503.03669.

[2] A. Almheiri and J. Polchinski. Models of AdS_2 backreaction and holography. *JHEP* 11 (2015) 014; arXiv:1402.6334.

[3] U. Bauer. Ripser: efficient computation of Vietoris–Rips persistence barcodes. *J. Appl. Comput. Topol.* 5 (2021) 391–423; arXiv:1908.02518.

[4] G. Carlsson. Topology and data. *Bull. Amer. Math. Soc.* 46 (2009) 255–308.

[5] D. Cohen-Steiner, H. Edelsbrunner and J. Harer. Stability of persistence diagrams. *Discrete Comput. Geom.* 37 (2007) 103–120.

[6] H. Edelsbrunner, D. Letscher and A. Zomorodian. Topological persistence and simplification. *Discrete Comput. Geom.* 28 (2002) 511–533.

[7] R. Jackiw. Lower dimensional gravity. *Nucl. Phys. B* 252 (1985) 343–356.

[8] A. Kitaev and S. J. Suh. The soft mode in the Sachdev–Ye–Kitaev model and its gravity dual. *JHEP* 05 (2018) 183; arXiv:1711.08467.

[9] J. Maldacena. Eternal black holes in anti-de Sitter. *JHEP* 04 (2003) 021; arXiv:hep-th/0106112.

[10] J. Maldacena and D. Stanford. Remarks on the Sachdev–Ye–Kitaev model. *Phys. Rev. D* 94 (2016) 106002; arXiv:1604.07818.

[11] J. Maldacena, D. Stanford and Z. Yang. Conformal symmetry and its breaking in two dimensional nearly anti-de Sitter space. *PTEP* 2016 (2016) 12C104; arXiv:1606.01857.

[12] J. Maldacena and L. Susskind. Cool horizons for entangled black holes. *Fortschr. Phys.* 61 (2013) 781–811; arXiv:1306.0533.

[13] T. G. Mertens, G. J. Turiaci and H. L. Verlinde. Solving the Schwarzian via the conformal bootstrap. *JHEP* 08 (2017) 136; arXiv:1705.08408.

[14] B. Olsthoorn. Persistent homology of quantum entanglement. *Phys. Rev. B* 107 (2023) 115174.

[15] S. Ryu and T. Takayanagi. Holographic derivation of entanglement entropy from AdS/CFT. *Phys. Rev. Lett.* 96 (2006) 181602; arXiv:hep-th/0603001.

[16] S. Sachdev and J. Ye. Gapless spin-fluid ground state in a random quantum Heisenberg magnet. *Phys. Rev. Lett.* 70 (1993) 3339–3342.

[17] C. Teitelboim. Gravitation and Hamiltonian structure in two spacetime dimensions. *Phys. Lett. B* 126 (1983) 41–45.

[18] A. Zomorodian and G. Carlsson. Computing persistent homology. *Discrete Comput. Geom.* 33 (2005) 249–274.